

Position-Taking from the Kernel Up: The Four-Cell Partition from Heyting Bedrock to Bach, Schoenberg, and Beckett

Chris Brink

FalseWork (falsework.dev)

*Version 0.1 (June 2026); pointer note July 2026. Expository synthesis. This note runs the four-position structure end to end — from the kernel-checked partition theorem at the bottom to the interpretive placement of works (Bach, Berg, Schoenberg, Beckett, and machine-generated process music) at the top — and its single discipline is to mark, at every rung, where the verified floor stops and interpretation begins. Status: the formal floor (§2–3) is **[K]**, kernel-checked in Lean 4 / Mathlib4, cited with artifacts. Everything from §5 onward — every placement of an actual work or composer in a cell — is **[A]**, a structural reading that does not inherit the floor’s certainty. The contribution of the note is the **map between the two**, and the honesty of the seam. The kernel decides what the cells are; it does not decide where Bach, Schoenberg, or Beckett sit in them. **Navigation (July 2026):** the programme front door is now `field-guide.md`; the deep music specimen is `bach-at-the-kernel.md` (v0.2, with opened-square landmarks). This note remains the seam-marked placement walk from bedrock to Beckett — not superseded, just no longer the first document to hand a new reader.*

1. What this note is, and the one rule it keeps

The framework’s central object is a four-cell partition of a Heyting algebra relative to a distinguished element (the *kernel*). The partition is a theorem. The use of that partition as a vocabulary for how artists and writers take positions toward inherited structure is an interpretation. This note connects the two — bedrock to Beckett — because the connection is the framework’s actual claim, and because a connection drawn carelessly is exactly the failure the project exists to prevent: an interpretive placement wearing the credential of a proof.

So the rule, stated once and kept throughout: **the formal floor is load-bearing for what the cells are; it is silent on where any actual work sits**. Section markers carry the tag. A reader who wants only the verified content can stop after §3; a reader who wants the reading should carry §3’s boundary with them into §5–7.

2. The bedrock (all [K], cited)

Fix a Heyting algebra $\mathfrak{H}$ — concretely, the lattice of subobjects of an object in an elementary topos, or a finite Heyting algebra. Fix a *kernel* element k . Define four conditions on an element x (equivalently, on the image of a morphism) relative to k :

- **Infrastructure** $\text{--- } x \hat{=} k$
- **Refusal** $\text{--- } x \hat{=} k^\perp$ (factors through the pseudo-complement)
- **Exploitation** $\text{--- } x \hat{=} k \hat{=} k^\perp \hat{=} k$ (inside the double-negation closure, exceeding the kernel)
- **Distribution** $\text{--- } x \hat{=} k \hat{=} k^\perp \hat{=} k$ (straddles both poles)

Theorem (four-position partition) [K]. These four conditions are disjoint and exhaustive over the morphism space (modulo the trivial-image edge case), against the in-repo HeytingAlgebra (Subobject $\mathcal{Y}$) instance. Lean: `four_position_partition, lattice_four_position_partition (Positions/Partition.lean)`; the subobject Heyting structure is upstreamed as Mathlib PR #39618. A **Commitment** gate operates as a binary fixedness condition *within* each cell, not as a fifth cell (the two-parameter unification of the four predicates is closed *negative* — they are propositional-shape-distinct, not specializations of one uniform term).

Three further kernel-checked facts give the partition a floor and a forcing:

- **The trichotomy [K]** (`allFourCellsInhabited_iff`): in *any* Heyting algebra all four cells are simultaneously inhabited at k **iff** $k \hat{=} k^\perp \hat{=} k$ — i.e. iff k is *ordinary* (neither regular nor dense, Citkin’s §2.1 term). The third clause, $k \hat{=} k^\perp \hat{=} k$, is the strict containment $k < k^\perp$ — **the formal content of the comma**. The Exploitation cell is non-empty exactly when the comma is strict.
- **The threshold [K]** (`ordinary_card_ge_six`): any Heyting algebra with an ordinary element has at least six elements. Below six, the partition is degenerate; $\mathbb{Z}_5$ carries no kernel.
- **The weld and the keystone [K]**: the music lattice `Div12` (subgroup lattice of $\hat{\mathbb{Z}}_{12}$, the pitch-class universe of twelve-tone equal temperament) is $\mathbb{Z}_6$, the smallest non-degenerate stage of the free Heyting algebra on one generator; the tritone is the generator/kernel. The all- n kernel law now holds **unconditionally** — the Nishimura enumeration it once cited [C] is itself kernel-checked (`generatedBy_isLadderValue, nishimura_kernel_unique_of_generated`; the Rieger–Nishimura normal form, June 2026).

That is the floor. It does not move under any interpretation below it.

3. The hinge: the kernel is the entry condition

Every cell predicate in §2 is defined *relative to* k . Remove k and the four conditions are not false — they are *undefined*. There is no “inside the kernel,” no “complement of the kernel,” no “residue of the kernel,” when there is no kernel.

This is the load-bearing structural fact for everything above, so state it sharply:

The partition is a classifier with a domain of definition. Its entry condition is the presence of a kernel. Material with no kernel does not fall into a cell — including no cell. It has not entered the structure at all.

And here is the single **[A]** bridge on which the entire interpretive edifice rests, named so it cannot hide: *the formal “presence of an ordinary kernel element” is read, in the music domain, as the presence of a **tonal center** — a home a listener can orient to.* This identification is not a theorem. It is the calibrating analogy, and every placement downstream inherits its status, not $\hat{\mathbb{H}}_2$'s.

Two senses of “kernel,” and where they meet. This note uses the word in two registers, and the honesty of everything below depends on not conflating them. The **formal kernel** is the ordinary element of $\hat{\mathbb{H}}_2$ — the object the partition theorem is *about*. The **working kernel** is what a practitioner takes a position toward: the inherited generative constraint a domain cannot escape (the four prose criteria — *prior, monogenic, inescapable, self-limiting* — of Paper 1). These are not two unrelated things sharing a word: in music they *provably coincide*. “Monogenic” is one-generated (`tritone_generates`); “self-limiting” is $\hat{\mathbb{A}}\text{-}\hat{\mathbb{A}}\text{-}k \hat{\mathbb{A}}\% k$, the comma (the ordinary condition itself); “inescapable” is the normal form — every element is a term in the generator (`generatedBy_isLadderValue`). So at the **Bach end** of this note the working kernel and the formal kernel are the same object, which is exactly why the music placements may ride the $\hat{\mathbb{H}}_2$ floor. But the coincidence is **earned, not free**: $\mathbb{H}_8(\text{unique_ordinary_converse_false})$ is a Heyting algebra that is *self-limiting* (a unique ordinary element) yet *not monogenic* (not one-generated by it) — proof that the two faces of “kernel” can come apart. So a domain may have a working kernel whose formal kernel has not been established, or does not exist. At the **Beckett end** of this note that is exactly the situation: literature has a working kernel (inherited form, to be refused) and *no* formal kernel at all. The full treatment of this seam — the four criteria, the $\mathbb{H}_8$ decoupling, the per-domain gradient — belongs in Paper 1 and `comma-formal-structure-note.md`; what matters here is that the word is doing two jobs, they meet only in music, and the placements inherit their floor accordingly.

One empirical check exists for the bridge, and it has been run (see `scripts/wolframtones-classification-experiment.py`, `wolframtones-adversarial-test.py`): cellular-automaton process music — structured but tonally centerless — is correctly held at the door. Standard key-finding reports no tonal center in raw CA output, and no *fair* (content-independent) relabeling of pitch-class space can manufacture one; only an imposed scale produces a center. So the entry condition does real gatekeeping: ordered-but-centerless input does not enter. This is a modest **[K]-adjacent** result about the bridge — not about the partition theorem — and it is the only falsifiable contact point in the whole stack.

4. The four cells in one breath (formal predicate $\hat{A}$ · interpretive gloss)

Cell	Formal (relative to k)	Interpretive gloss [A]
Infrastructure	$x \hat{a} \approx k$	builds on the kernel as transparent ground
Distribution	straddles k and $k \acute{a} \mathfrak{J}$	encounters the gap and redistributes it across sites
Exploitation	$x \hat{a} \approx k \acute{a} \mathfrak{J} \acute{a} \mathfrak{J}$, $x \hat{a} \mathfrak{S},, k$ (the residue $k \acute{a} \mathfrak{J} \acute{a} \mathfrak{J} \hat{a}^- - k$)	inhabits the kernel's shadow — the comma — that the kernel can't reach
Refusal	factors through $k \acute{a} \mathfrak{J}$	structured <i>against</i> the kernel; rejects its authority
(exit)	<i>no k</i>	pre-kernel; has not entered — not a cell

Two structural facts about this table carry the whole reading, and both are consequences of the *formal* definitions, not of taste:

1. **Exploitation is the most kernel-dependent cell.** It exists only when $k \acute{a} \mathfrak{J} \acute{a} \mathfrak{J} \hat{a} \approx k$ — only when the kernel is genuinely non-Boolean. It is the comma made into a place.
2. **Refusal is kernel-bound, not kernel-free.** $k \acute{a} \mathfrak{J}$ is the largest element disjoint from k — it is as far from the kernel as one can get *while remaining entirely defined by it*. Refusal is the maximal distance from the kernel that is still a position relative to the kernel. You do not leave the domain by refusing; refusal is the most charged way of staying in it.

5. The placements — bottom to top (all [A])

Each placement is reasoned from the cell's *formal* predicate down to a *work*, so the logic is visible and the leap is locatable. The leap is always at the same joint: from “this element of the lattice satisfies the predicate” to “this piece of music or literature does.”

- **Bach — Infrastructure.** *Es ist genug* (BWV 60) states the kernel — the tritone — inside the functional ground itself: the chorale's incipit outlines the tritone and the surrounding tonal machinery absorbs it as substrate. The kernel is present and worked *directly*, $x \hat{a} \approx k$. The structure is stated from inside the home.
- **Berg — Exploitation.** The atonalist who would not release the kernel. Berg builds triads into his tone rows and, in the Violin Concerto, literally quotes *Es ist genug* and ends the row on the whole-tone tetrachord. Twelve-tone *technique*, but the tonal kernel is kept in view and worked as *residue* — $x \hat{a} \approx k \acute{a} \mathfrak{J} \acute{a} \mathfrak{J}$, $x \hat{a} \mathfrak{S},, k$. He inhabits the comma. The diminished-seventh / whole-tone material of the Exploitation cell is, almost literally, his idiom.
- **Schoenberg — Refusal.** “Emancipation of the dissonance” is liberation *from* the center, and the thing one is freed *from* remains the reference point. Schoenberg's atonality is *programmatic negation*: structured by the rejection of the kernel, $x \hat{a} \approx k \acute{a} \mathfrak{J}$. He is more atonal than Berg — and in the lens that makes him *more Refusal*, deeper into $k \acute{a} \mathfrak{J}$, not closer to any exit. (True to the algebra: he kept sliding back — the tonal returns, the motivic tonics-of-form. The kernel never left; he was refusing it, not without it.)

- **Beckett — Refusal at its limit (literary).** The stripping-toward-silence, toward the unword-able void. Beckett strains for the exit — the kernel-less nothing — and never reaches it: every attempt to reach silence *produces more structured language*. “I can’t go on, I’ll go on” **is the comma**: the non-closure, the residue that will not resolve to the terminal silence it longs for. He does not exit; he writes the impossibility of exiting. He is the purest depiction of why there is no door out *for one who wants one*.
- **Process music — the exit (doesn’t enter).** Cellular-automaton and stochastic music with no referential center: not *refusing* a kernel, simply *having* none. Pre-kernel. The Â§3 experiment is its worked instance.

6. The one non-obvious law the placements obey

The placements are not a ladder from “tonal” to “atonal” with the exit at the far end. The structural ordering, forced by Â§4’s two facts, is different and sharper:

***You exit the domain by indifference, not by intensity.** Refusal is the second-most kernel-charged cell. The modern movement toward originality — the drive to escape inherited bounded structure — was a Refusal gesture, and Refusal is *káŕe*, defined by what it rejects. The harder modernism pushed against the structure, the more tightly the structure defined it. It could never fully exit, because its energy was the wish to transcend the kernel, and that wish is kernel-bound. One leaves only when the kernel ceases to matter — which is not rupture but indifference, the move toward pure system or process (and, at the far end, the machine’s centerless output).*

This is why Schoenberg and Beckett *could not fully exit* and the cellular automaton *already has*: the automaton never wanted a center; they could not stop wanting to escape one.

7. The seam, and the honest account of what this is

Everything in Â§2 is kernel-checked and cited. Everything from Â§5 is [A]. The map between them — Â§3–4, 6 — is the note’s contribution, and it is itself [A]: a reasoned correspondence, not a theorem. Three honesties keep it from overreaching:

- **Literature has only the working kernel, not the formal one.** The music placements sit above a real formal object (Div12, the subobject lattice), where — per Â§3 — the working kernel and the formal kernel provably coincide. The Beckett placement does not: literature has a *working* kernel (inherited form, there to be refused) but **no formal kernel** — no kernel-bearing lattice of literary works. Applying *Refusal* to Beckett is transporting a vocabulary the theorem grounds *in music* to a domain where only the prose face of the kernel is present — metaphor calibrated by the music case, nothing more. The further from Div12, the thinner the floor; over literature there is no formal floor at all, only the calibrated word. (And by the ð8 decoupling of Â§3, this is not a mere “not yet formalized” — a working kernel need not have a formal kernel behind it.)
- **It converges with prior art, which is corroboration, not proof.** “The work is bound to what it negates” is Adorno’s *determinate negation*; “the artist cannot escape the precursor even in swerving” is Bloom’s *anxiety of influence* (relocated structurally in Paper 6). Three vocabu-

laries landing on the same shape is evidence the shape is real — but it means the contribution is the *unification* (music’s Refusal and literature’s anxiety-of-influence as the same *káŋæ* move), not the observation.

- **One move explains everything here — which is the exact double edge.** A single structural fact (Refusal is kernel-bound) just placed tonal music, Berg, Schoenberg, the modern movement, and Beckett. That is either depth or elasticity, and the two are not yet distinguished. A good lens fits everything in its domain; fitting is not evidence. The only place the structure has been made to *risk* a verdict it could fail is the CA entry-condition test (Â§3) — and there it survived. The placements above have not been made falsifiable, and the note does not pretend they have. What would make one of them checkable — a prediction about a work that the reading could get *wrong* — is the open problem this exposition hands forward.

The kernel-checked claim is small and exact: there is a four-cell partition of a Heyting algebra relative to an ordinary element, and the music lattice is its smallest forced instance. The reading is large and [A]: that this partition is the structure of position-taking toward inherited form, and that Bach, Berg, Schoenberg, and Beckett occupy four distinct relations to a single kernel they all, in their different ways, cannot get out from under. The first sentence is proved. The second is offered. This note exists so the line between them is never blurred — which is the only condition under which offering the second is honest.

Formal *record:* *Lean/FalseWorkPapers/Positions/Partition.Lean,* *Examples/*
NishimuraNormalForm.Lean, Examples/LadderCore.Lean, Examples/DivisorLattice12.Lean; experiments
at scripts/wolframtones-classification-experiment.py, scripts/wolframtones-adversarial-test.py;
companions papers/comma-formal-structure-note.md (the cell predicates in full), papers/bach-at-
the-kernel.md (the single-specimen reading), papers/four-position-exposition.md (the formal
exposition), papers/connecting-the-spine.md (the spine). Musical references: Bach, Cantata BWV
60, chorale Es ist genug (Ahle); Berg, Violin Concerto (1935); Schoenberg, the free-atonal and
twelve-tone works; Beckett, The Unnamable (1953), Worstward Ho (1983). Theoretical kin:
Adorno (determinate negation), Bloom (anxiety of influence).